

Building and Using Custom Terraform Modules

```
seife@SeifEddine:~/projects/terraform/modules/payroll$ ls
seife@SeifEddine:~/projects/terraform/modules/payroll$ pwd
/home/seife/projects/terraform/modules/payroll
seife@SeifEddine:~/projects/terraform/modules/payroll$ touch main.tf variables.tf output
s.tf
seife@SeifEddine:~/projects/terraform/modules/payroll$ nano main.tf variables.tf outputs
.tf
seife@SeifEddine:~/projects/terraform/modules/payroll$ nano variables.tf
seife@SeifEddine:~/projects/terraform/modules/payroll$ cd ../../
pwd
ls
/home/seife/projects/terraform
modules
seife@SeifEddine:~/projects/terraform$
```

```
seife@SeifEddine:~/projects/terraform$ aws configure
AWS Access Key ID [*****BQ7K]:
AWS Secret Access Key [*****8bGx]:
Default region name [eu-west-3]: eu-central-1
Default output format [text]:
seife@SeifEddine:~/projects/terraform$ nano main.tf
seife@SeifEddine:~/projects/terraform$ terraform init
Initializing modules...
Initializing provider plugins found in the configuration...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Using previously-installed hashicorp/aws v6.51.0
```

Initializing the backend...

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
seife@SeifEddine:~/projects/terraform$
```

```
seife@SeifEddine:~/projects/terraform$ terraform apply -auto-approve
module.payroll.data.aws_ami.ubuntu: Reading...
module.payroll.aws_dynamodb_table.payroll_db: Refreshing state... [id=PayrollDB]
module.payroll.aws_s3_bucket.payroll_docs: Refreshing state... [id=payroll-docs-seife-20260623]
module.payroll.data.aws_ami.ubuntu: Read complete after 0s [id=ami-042dc8681de073ac4]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:

+ create

Terraform will perform the following actions:

```
# module.payroll.aws_instance.payroll_server will be created
+ resource "aws_instance" "payroll_server" {
  + ami                  = "ami-042dc8681de073ac4"
  + arn                  = (known after apply)
  + associate_public_ip_address = (known after apply)
  + availability_zone     = (known after apply)
  + disable_api_stop      = (known after apply)
  + disable_api_termination = (known after apply)
  + ebs_optimized         = (known after apply)
  + enable_primary_ipv6   = (known after apply)
  + force_destroy         = false
  + get_password_data     = false
  + host_id               = (known after apply)
  + host_resource_group_arn = (known after apply)
  + iam_instance_profile   = (known after apply)
  + id                   = (known after apply)
  + instance_initiated_shutdown_behavior = (known after apply)
  + instance_lifecycle    = (known after apply)
  + instance_state        = (known after apply)
  + instance_type         = "t2.micro"
  + ipv6_address_count    = (known after apply)
  + ipv6_addresses        = (known after apply)
  + key_name              = (known after apply)
```

0 0

Sign In

```
- versioning {
  - enabled    = false -> null
  - mfa_delete = false -> null
}
```

Plan: 0 to add, 0 to change, 3 to destroy.

```
module.payroll.aws_s3_bucket.payroll_docs: Destroying... [id=payroll-docs-seife-20260623]
module.payroll.aws_instance.payroll_server: Destroying... [id=i-075de62c7089073bc]
module.payroll.aws_dynamodb_table.payroll_db: Destroying... [id=PayrollDB]
module.payroll.aws_s3_bucket.payroll_docs: Destruction complete after 0s
module.payroll.aws_dynamodb_table.payroll_db: Destruction complete after 6s
module.payroll.aws_instance.payroll_server: Still destroying... [id=i-075de62c7089073bc, 00m10s elapsed]
module.payroll.aws_instance.payroll_server: Still destroying... [id=i-075de62c7089073bc, 00m20s elapsed]
module.payroll.aws_instance.payroll_server: Still destroying... [id=i-075de62c7089073bc, 00m30s elapsed]
module.payroll.aws_instance.payroll_server: Destruction complete after 30s
```

Destroy complete! Resources: 3 destroyed.

```
seife@SeifEddine:~/projects/terraform$
```